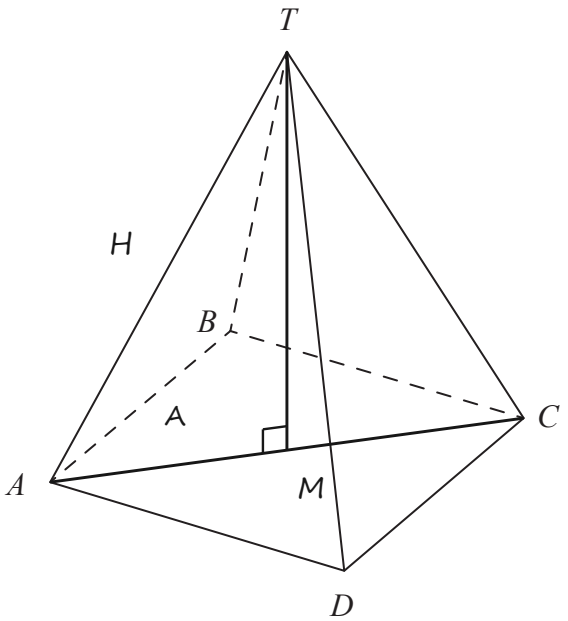
12 Here is a pyramid with a square base $ABCD$.



TM = 12 m (O)

$AB = 5\,\text{m}$

The vertex T is 12 m vertically above the midpoint of AC.

Calculate the size of angle TAC.

Base diagonal: $AC^2 = 5^2 + 5^2 = 50$

1 mark
Use Pythagoras in the square base.

$AM = AC \div 2 = \sqrt{50} \div 2 = 3.535533906...$

1 mark
The midpoint halves the diagonal.

$$\tan TAC = \frac{12}{\sqrt{50} \div 2}$$

1 mark
Opposite ÷ adjacent in triangle TAM.

$TAC = \tan^{-1}(12 \div (\sqrt{50} \div 2))$

$= 73.58355963...° = 73.6°$

1 mark
Correct angle.

°

(Total for Question 12 is 4 marks)